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Dustproof member, camera module having the dustproof member and electronic device having the camera module

Abstract

A dustproof member includes a base plate, a dustproof plate, and a connecting portion. A first through hole is defined by an inner surface of the base plate. A second through hole is defined by an inner surface of the dustproof plate. The connecting portion has a shape of a hollow barrel and defines a hollow cavity. Two opposite ends of the connecting portion are respectively connected to an end of the inner surface of the base plate and an end of the inner surface of the dustproof plate, thereby forming a dust collecting groove between the base plate and the dustproof plate. A camera module having the dustproof member and an electronic device having the camera module are also provided.

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Background/Summary

FIELD
(1) The subject matter herein generally relates to a dustproof member, a camera module having the dustproof member, and an electronic device having the camera module.
BACKGROUND
(2) Most camera modules include a lens, a voice coil motor, a filter, a photosensitive chip, a holder, and a circuit board. The lens is installed in the voice coil motor, and a gap is formed between the lens and the voice coil motor. When dust and impurities enter from the gap and fall on the filter, the image becomes stained, and the dust and impurities will also rise and fall when the camera module is shaken or dropped.
(3) Therefore, there is room for improvement within the art.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Implementations of the present disclosure will now be described, by way of embodiments, with reference to the attached figures.

(2) FIG. **1** is a diagram of an embodiment of a camera module according to the present disclosure.

(3) FIG. **2** is an exploded, diagrammatic view of an embodiment of a camera module according to the present disclosure.

(4) FIG. **3** is an exploded, diagrammatic view of an embodiment of a part of a camera module of according to the present disclosure.

(5) FIG. **4** is a cross-sectional view of the camera module taken along IV-IV line of FIG. **1**.

(6) FIG. **5** is an exploded, diagrammatic view of another embodiment of a part of a camera module according to the present disclosure.

(7) FIG. **6** is a diagram of an embodiment of an electronic device according to the present disclosure.

DETAILED DESCRIPTION

(8) It will be appreciated that for simplicity and clarity of illustration, where appropriate, reference numerals have been repeated among the different figures to indicate corresponding or analogous elements. In addition, numerous specific details are set forth in order to provide a thorough understanding of the embodiments described herein. However, it will be understood by those of ordinary skill in the art that the embodiments described herein can be practiced without these specific details. In other instances, methods, procedures, and components have not been described in detail so as not to obscure the related relevant feature being described. Also, the description is not to be considered as limiting the scope of the embodiments described herein. The drawings are not necessarily to scale, and the proportions of certain parts may be exaggerated to better illustrate details and features of the present disclosure.

(9) The disclosure is illustrated by way of example and not by way of limitation in the figures of the accompanying drawings, in which like references indicate similar elements. It should be noted that references to “an” or “one” embodiment in this disclosure are not necessarily to the same embodiment, and such references mean “at least one.”

(10) The term “comprising,” when utilized, means “including, but not necessarily limited to”; it specifically indicates open-ended inclusion or membership in the so-described combination, group, series, and the like.

(11) FIG. **1** illustrates a first embodiment of a camera module **100**. Referring to FIGS. **1** and **2**, the camera module **100** includes a circuit board **10**, a photosensitive chip **20**, a holder **30**, a filter **40**, a voice coil motor **50**, a dustproof member **60**, and a lens assembly **70**. The photosensitive chip **20** is arranged on a surface of the circuit board **10**. The holder **30** is mounted on the surface of the circuit board **10**. The voice coil motor **50** is located at a side of the holder **30** facing away from the circuit board **10**. The filter **40** is located at a side of the photosensitive chip **20** facing away from the circuit board **10** and is spaced from the photosensitive chip **20**. The filter **40** is mounted in the holder **30** and corresponds to the photosensitive chip **20**. The dustproof member **60** is arranged on a surface of the voice coil motor **50** facing away from the holder **30**. A part of the lens assembly **5** is received in the voice coil motor **50**.

(12) In at least one embodiment, the circuit board **10** may be a flexible circuit board, a rigid circuit board, or a flexible-rigid circuit board. Specifically, the circuit board **10** is a flexible-rigid circuit board, and includes a first rigid portion **101**, a second rigid portion **102**, and a flexible portion **103**. The flexible portion **103** is connected to the first rigid portion **101** and the second rigid portion **102**.

(13) The circuit board **10** may further include an electrical connecting portion **11** mounted on the second rigid portion **102**. When the camera module **100** is applied in an electronic device (such as

mobile phone), the electrical connecting portion **11** is used to implement signal transmission between the camera module **100** and other components of the electronic device.

(14) The circuit board **10** may further include a first reinforcement plate **12** and a second reinforcement plate **13**. The first reinforcement plate **12** is mounted on a surface of the second rigid portion **102** facing away from the electrical connecting portion **11**, the second reinforcement plate **13** is mounted on a surface of the first rigid portion **101** facing away from the holder **30**. In at least one embodiment, a material of the first reinforcement plate **12** and the second reinforcement plate **13** are both metal (such as stainless steel).

(15) The photosensitive chip **20** may be mounted on the first rigid portion **101** by a first adhesive layer **21**.

(16) The holder **30** may be fixed on the first rigid portion **101** by a second adhesive layer **23**. The holder **30** may be substantially a hollow rectangular structure and includes a first surface **301** and a second surface **302** facing away from the first surface **301**. The first surface **301** faces the voice coil motor **50**, the second surface **302** facing the circuit board **10**. A third through hole **31** is defined in the holder **30** and penetrates the first surface **301** and the second surface **302**. A groove **32** is recessed inward from the first surface **301**, and the groove **32** surrounds and communicates with the third through hole **31**. The filter **40** is received in the groove **32**.

(17) In at least one embodiment, the filter **40** may be substantially rectangular. The filter **40** is fixed in the holder **30** by a third adhesive layer **42**.

(18) The voice coil motor **50** may be substantially a hollow rectangular structure. A receiving hole **52** penetrates the voice coil motor **50**. A part of the lens assembly **70** is received in the receiving hole **52**. The voice coil motor **50** is fixed on the holder **30** by a fourth adhesive layer **54**. The voice coil motor **50** may be made of metal or plastic.

(19) Referring to FIGS. **2**, **3**, and **4**, the dustproof member **60** includes a base plate **611**, a dustproof plate **613**, and a connecting portion **614**. An inner surface of the base plate **611** surrounds to form a first through hole **602**, and an inner surface of the dustproof plate **613** surrounds to form a second through hole **604**. The connecting portion **614** is in a shape of a hollow barrel and has a hollow cavity **607**. Two opposite ends of the connecting portion **614** are respectively connected to an end of the inner surface of the base plate **611** and an end of the inner surface of the dustproof plate **613**.

(20) In at least one embodiment, the base plate **611** may be substantially rectangular, the first through hole **602** may be substantially circular, the dustproof plate **613** may be substantially annular.

(21) A surface of the dustproof plate **613** facing away from the base plate **611** is recessed inward to form a dustproof groove **603** for containing dust particles. The dustproof groove **603** may be annular and arranged around the second through hole. A dustproof glue may be pasted in the dustproof groove **603** for attracting the dust particles.

(22) The base plate **611**, the dustproof plate **613**, and the connecting portion **614** cooperate to form a dust collecting groove **605** (shown in FIG. **4**). The dust collecting groove **605** is located between the base plate **611** and the dustproof plate **613**. In at least one embodiment, a glue may be pasted in the dust collecting groove **605** to attract dust particles.

(23) A side of the base plate **611** facing away from the dustproof plate **613** is connected to the voice coil motor **50**. The first through hole **602** corresponds to the receiving hole **52**. In at least one embodiment, a diameter of the first through hole **602** is equal to or less than a diameter of the receiving hole **52**. A plurality of positioning holes **606** penetrate the base plate **601** for positioning the base plate **601** on the voice coil motor **50**.

(24) In at least one embodiment, the connecting portion **614** surrounds along a central axis G to define the hollow cavity **607**. Along a direction perpendicular to the central axis a diameter of the hollow cavity **607** gradually decreases from the base plate **611** to the dustproof plate **613**, so as to further prevent dust from entering.

(25) Referring to FIGS. **2** and **4**, the dustproof member **60** may be fixed on the voice coil motor **50**

by a fifth adhesive layer **62**.
(26) The lens assembly **70** may include a main body **71** and a dust cover **72**. A portion of the main body **71** is received in the receiving hole **52**. The dust cover **72** is arranged on an end of the main body **71** facing away from the voice coil motor **50**.
(27) The main body **71** is formed by at least one lens portion. In at least one embodiment, the main body **71** may include a first lens portion **701**, a second lens portion **702**, a third lens portion **703**, and a fourth lens portion **704**. The first lens portion **701**, the second lens portion **702**, the third lens portion **703**, and the fourth lens portion **704** are connected in that sequence. The dust cover **72** is located at a side of the first lens portion **701** facing away from the second lens portion **702**. In at least one embodiment, a diameter of the first lens portion **701**, a diameter of the second lens portion **702**, a diameter of the third lens portion **703**, and a diameter of the fourth lens portion **704** may increase in that sequence. The first lens portion **701**, the second lens portion **702**, the third lens portion **703**, and the fourth lens portion **704** may be integrally molded to form the main body **71**.
(28) FIG. **5** illustrates a second embodiment of a part of a camera module **200**. The camera module **200** is different from the camera module **100** in that the dustproof member **60** is divided into at least two sub-dustproof parts **60***a* along the central axis G Each of the sub-dustproof parts **60***a* includes a part of the base plate **611**, a part of the dustproof plate **613**, and a part of the connecting portion **614**.
(29) FIG. **6** illustrates an embodiment of an electronic device **300** including the above camera module **100** or **200**. The electronic device **300** may be, but not limited to, a mobile phone, a wearable device, a computer device, a vehicle or a monitoring device.
(30) It is to be understood, even though information and advantages of the present embodiments have been set forth in the foregoing description, together with details of the structures and functions of the present embodiments, the disclosure is illustrative only; changes may be made in detail, especially in matters of shape, size, and arrangement of parts within the principles of the present embodiments to the full extent indicated by the plain meaning of the terms in which the appended claims are expressed.

Claims

1. A camera module comprising: a voice coil motor comprising a receiving hole; a lens assembly; and a dustproof member comprising: a base plate, wherein a first through hole is defined by an inner surface of the base plate; a dustproof plate, wherein a second through hole is defined by an inner surface of the dustproof plate; and a connecting portion; wherein the connecting portion has a shape of a hollow barrel and defines a hollow cavity, two opposite ends of the connecting portion are respectively connected to an end of the inner surface of the base plate and an end of the inner surface of the dustproof plate, thereby forming a dust collecting groove between the base plate and the dustproof plate; the lens assembly sequentially extends through the first through hole, the hollow cavity and the second through hole, a part of the lens assembly is located on a side of the dustproof plate facing away from the base plate, the voice coil motor is located on a side of the base plate facing away from the dustproof plate, a part of the lens assembly is received in the receiving hole, a surface of the base plate facing away from the dustproof plate is connected to the voice coil motor, the first through hole corresponds to the receiving hole; the connecting portion surrounds along a central axis to define the hollow cavity, along a direction perpendicular to the central axis, a diameter of the hollow cavity gradually decreases from the base plate to the dustproof plate.
2. The camera module of claim 1, wherein a surface of the dustproof plate facing away from the base plate is recessed inward to form a dustproof groove.
3. The camera module of claim 2, wherein a dustproof glue is pasted in the dustproof groove.
4. The camera module of claim 1, wherein the connecting portion surrounds along a central axis to

define the hollow cavity, along a direction perpendicular to the central axis, a diameter of the hollow cavity linearly decreases from the base plate to the dustproof plate.

5. The camera module of claim 1, wherein the connecting portion surrounds along a central axis to define the hollow cavity, the dustproof member is divided into at least two sub-dustproof parts along the central axis, each of the at least two sub-dustproof parts includes a part of the base plate, a part of the dustproof plate, and a part of the connecting portion.

6. The camera module of claim 1, wherein along a same direction, a diameter of the first through hole is equal to or less than a diameter of the receiving hole.

7. An electronic device comprising: a camera module comprising: a voice coil motor comprising a receiving hole; a lens assembly; and a dustproof member comprising: a base plate, wherein a first through hole is defined by an inner surface of the base plate; a dustproof plate, wherein a second through hole is defined by an inner surface of the dustproof plate; and a connecting portion; wherein the connecting portion has a shape of a hollow barrel and defines a hollow cavity, two opposite ends of the connecting portion are respectively connected to an end of the inner surface of the base plate and an end of the inner surface of the dustproof plate, thereby forming a dust collecting groove between the base plate and the dustproof plate; the lens assembly sequentially extends through the first through hole, the hollow cavity and the second through hole, a part of the lens assembly is located on a side of the dustproof plate facing away from the base plate, the voice coil motor is located on a side of the base plate facing away from the dustproof plate, a part of the lens assembly is received in the receiving hole, a surface of the base plate facing away from the dustproof plate is connected to the voice coil motor, the first through hole corresponds to the receiving hole; the connecting portion surrounds along a central axis to define the hollow cavity, along a direction perpendicular to the central axis, a diameter of the hollow cavity gradually decreases from the base plate to the dustproof plate.

8. The electronic device of claim 7, wherein a surface of the dustproof plate facing away from the base plate is recessed inward to form a dustproof groove.

9. The electronic device of claim 8, wherein a dustproof glue is pasted in the dustproof groove.

10. The electronic device of claim 7, wherein the connecting portion surrounds along a central axis to define the hollow cavity, along a direction perpendicular to the central axis, a diameter of the hollow cavity linearly decreases from the base plate to the dustproof plate.

11. The electronic device of claim 7, wherein the connecting portion surrounds along a central axis to define the hollow cavity, the dustproof member is divided into at least two sub-dustproof parts along the central axis, each of the at least two sub-dustproof parts includes a part of the base plate, a part of the dustproof plate, and a part of the connecting portion.

12. The electronic device of claim 7, wherein along a same direction, a diameter of the first through hole is equal to or less than a diameter of the receiving hole.